From Green-hyperbolic complexes to conformal detours

Bridge note for Lorentzian PDE, Green-hyperbolic-complex, and BGG/detour geometry researchers. Status: abstract transfer theorem certified with Berger, flat-Minkowski, and curved Nariai consumers, 18 July 2026.

1. Their object and the unresolved question

Benini, Musante, and Schenkel define a Green-hyperbolic complex through retarded and advanced Green homotopies, prove uniqueness up to a contractible space of choices, identify the causal quasi-isomorphism, and compare covariant and fixed-time Poisson structures up to homotopy (arXiv:2207.04069). Gover, Somberg, and Souček construct formally self-adjoint Yang–Mills detour complexes and obtain conformal BGG-related complexes from tractor connections on Bach-flat four-manifolds (arXiv:math/0606401). Neither scope is being challenged here.

Our question is constructive: if a hyperbolic tractor/prolonged parent is reduced to a fourth-order detour complex by differential contractions, homological perturbation, and finite support-local shears, what hypotheses ensure that retarded/advanced homotopies, cyclic adjoints, and current pairings descend together?

```
hyperbolic tractor parent       support-local cyclic SDR       detour/Bach complex
       15/60/60/15         ------------------------------>        4/9/9/4
             |                                                       |
       Green homotopies                                      transferred Green
       and parent current                                     and current pairing
             +------------------ exact comparison ----------------+
```

2. Exact dictionary

Item	Green-hyperbolic/detour language	This project	Conversion or caveat
Differential	cochain differential	classical BV–BFV differential q_1	Degree and sign conventions are fixed by the odd pairing.
Propagation	retarded/advanced Green homotopy	$\Lambda_\pm$ on compact or spacelike-compact support	These are chain homotopies, not an inverse of the isolated Bach operator.
Reduction	quasi-isomorphism or contraction	cyclic differential SDR plus curved HPL and shears	Every shear is finite order and support-local.
Geometric parent	bundle complex with connection	adjoint-tractor/Yang–Mills parent and curvature prolongation	The concrete theorem uses conformal flatness of the cylinder.
Reduced object	Green-hyperbolic complex	metric Bach detour/BV complex	The reduced central operator need not have scalar principal symbol.
Pairing	differential pairing and Poisson structures	odd BV pairing, Green current, and residual pairing	Equality is transported up to the declared chain homotopy.
Boundaries	globally hyperbolic support categories	$\mathbb{R} \times S^3$, no timelike boundary	Boundary flux is a separate theorem.

3. Reproduced benchmark

The shared benchmark is the Green-homotopy identity and its causal difference. On the complete 386-row prolonged cylinder complex, the repository verifies coefficientwise that the retarded and advanced degree-minus-one maps satisfy the chain-homotopy equations on the declared support categories. Their difference induces the compact-to-global quasi-isomorphism, and temporal cutoffs recover the fifteen conformal-Killing endpoint classes without duplication. The direct current calculation transports the normalized H^4 pairing to I_2.

This is the concrete analogue of the abstract causal quasi-isomorphism and Poisson-compatibility statements. The benchmark is reproduced by:

```
python3 symbolic/verify_conformal_final_covariant_transport.py
python3 symbolic/verify_conformal_covariant_H4_proof_ledger.py --check --guards
```

The exact claim and release rail are documented in Paper 8 and its reproduction guide.

4. Added result

Abstract cyclic causal-transfer theorem. Let (C, q_C) support-locally and cyclically deformation-retract onto (E, q_E) through (i, p, h). If E has advanced and retarded Green homotopies, then

$$\Lambda_{C,\pm} = h + i\Lambda_{E,\pm}p$$

are advanced and retarded Green homotopies on C. Their same-sided support and complementary-degree cyclic adjoint relation are preserved. The property is also closed under finite direct sums and finite-order support-local cyclic shears with finite-order inverses.

The proof is algebraic once endpoint propagation exists:

$$q_C\Lambda_{C,\pm} + \Lambda_{C,\pm}q_C = (1 - ip) + i(q_E\Lambda_{E,\pm} + \Lambda_{E,\pm}q_E)p = 1.$$

Support follows because i, p, h do not enlarge support. Taking the graded adjoint proves advanced/retarded reversal from $i^\sharp = p$ and the pairing-derived degreewise sign involutions:

$$h^\sharp = \Sigma_C h \Sigma_C^{-1}, \qquad \Lambda_{E,+}^\sharp = \Sigma_E \Lambda_{E,-} \Sigma_E^{-1}.$$

Thus no uniform scalar sign is assumed across all degrees.

The same cyclic SDR also transfers an already constructed parent homotopy downward:

$$\Lambda_{E,\pm} = p\Lambda_{C,\pm}i, \qquad q_E\Lambda_{E,\pm} + \Lambda_{E,\pm}q_E = p(q_C\Lambda_{C,\pm} + \Lambda_{C,\pm}q_C)i = 1_E.$$

This is the direction used by differential tractor/BGG compression.

The content-addressed first consumer is the Berger gravity–clock complex:

```
54 = 28 algebraic + 26 causal,
Lambda54,+/- = S_cl + iota_cl Lambda26,+/- pi_cl.

64 = 28 algebraic + (26 gravity-clock + 10 Maxwell),
Lambda64,+/-
  = S64 + iota64 (Lambda26,+/- direct-sum LambdaM,+/-) pi64.
```

The exact certificate and independent replay are `ABSTRACT_CYCLIC_CAUSAL_TRANSFER`. New applications must first satisfy the strict **`consumer contract`**, which requires typed operator domains, boundary conditions, pairing-derived sign data, the exact cyclic SDR, causal-input Green data, and finite local inverses for all shears. The accepted Berger adapter is `BERGER_ABSTRACT_CAUSAL_TRANSFER_CONSUMER`. The theorem is background-uniform as a conditional statement. Three scoped `G2` consumers are now certified: the Berger lift above, a non-cylinder parent-to-endpoint descent on Minkowski, and a curved Nariai parent-detour-to-metric descent. The Minkowski consumer doubles the flat adjoint-tractor detour with opposite normalization, applies a cyclic triangular flavor shear, and descends the parent Hodge homotopy through the exact flat differential BGG retract. Its proof and portable adapter are `MINKOWSKI_DOUBLED_ADJOINT_TRACTOR_MIXED_DETOUR` and `MINKOWSKI_DOUBLED_ADJOINT_TRACTOR_CAUSAL_TRANSFER_CONSUMER`. The mixed unary presentation has a nonzero off-diagonal block, but is linearly equivalent to two free copies; it is a portability test, not an interacting model.

The Nariai consumer is the first substantial curved portability test. Its repaired ten-block parent-detour cone has degree ranks `15/140/140/15`, hence 310 components in total, and retracts cyclically and support-locally onto an independently certified 26-component metric Bach endpoint. Exact operator-polynomial replay verifies all ten blocks,

```
Lambda_310,+/- = H + I Lambda_metric,+/- P,
Q_310 Lambda_310,+/- + Lambda_310,+/- Q_310 = 1_310,
P Lambda_310,+/- I = Lambda_metric,+/-.
```

The advanced/retarded support and complementary-degree adjoint relation are preserved. The certificate and independent replay are `NARIAI_REPAIRED_310_ALL_ROW_GREEN_TRANSFER_V1` and its result-boundary report. This closes one curved single-background `G2` gate; it is not a uniform open-family theorem. No open `G3` background family has yet been proved to satisfy the analytic hypotheses uniformly.

The earlier conformal-cylinder construction remains the motivating detour example:

> **Concrete transfer theorem.** On the declared conformal-cylinder complex, the support-local cyclic contraction from the tractor/prolonged parent to the metric detour complex transfers retarded and advanced Green homotopies, their support properties, the causal quasi-isomorphism, and the current pairing. No canonical Green inverse of the isolated fourth-order metric operator is required.

Its proof factors the construction into curvature prolongation, exact tractor-to-BGG contraction, filtered homological perturbation, finite local shears, and cyclic adjoint transport. Each factor has an explicit inverse or homotopy identity and a support ledger. Recasting the full curved-cylinder construction as a standalone content-addressed consumer remains separate from the flat Minkowski portability pilot.

5. Consequence in their language

The example supplies a large, coefficient-level fourth-order gauge complex in which Green hyperbolicity belongs to the complex even though a preferred same-bundle scalar-principal factorization of its middle operator is excluded. For detour geometry, it adds a Lorentzian propagation layer to the tractor/BGG construction. For Green-hyperbolic complexes, it suggests a practical closure criterion under cyclic differential SDRs rather than only an existence definition.

6. Scope boundary

The abstract result is conditional: it does **not** establish endpoint Green hyperbolicity on a proposed background. It does not cover timelike boundaries, interactions, a Hadamard state, time-ordered products, or the quantum master equation. The surviving H^4 classes are centered deformation classes, not one-particle gravitons. Flat Minkowski and curved Nariai now supply two non-Berger consumers, but uniform `G3` background dependence and timelike-boundary domains remain open.

7. One useful question for adjacent experts

> Can the same consumer contract be verified uniformly on a nontrivial open family of curved backgrounds, including stable operator domains and degreewise Green estimates?

Reproducibility receipt

```
source papers: arXiv:2207.04069v2; arXiv:math/0606401v2
project source: Paper 8 artifact-ready snapshot and current master
verification: commands in section 3
dependency tag: LORENTZIAN-CAUSAL
generality level: ABSTRACT_CONDITIONAL_THEOREM; THREE_SCOPED_G2_CONSUMERS
lifecycle state: ABSTRACT_THEOREM_AND_THREE_SCOPED_CONSUMERS_CERTIFIED
claim flag: ABSTRACT_CAUSAL_TRANSFER_CERTIFIED
open fields: G3 family; boundary version; Hadamard transfer
```